A pipeline for Venusian surface compositional study using VIRTIS data at near IR spectral windows with most surface contribution.

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in Partial Fulfilment of the Requirements
for the Degree of

MASTER OF SCIENCE

by

Summit Bikram Nayak



to the

School of Physical Sciences

National Institute of Science Education and Research

Bhubaneswar

DEDICATION

I dedicate the M. Sc. to my beloved Family, Friends, and Lab mates who have been a continuous source of inspiration for me throughout the downs of my life in here, and for the ups i don't need any inspiration. Special thanks to the small and cute family over here, they have parts of my soul as Horcrux.

DECLARATION

I hereby declare that I am the sole author of this thesis in partial fulfillment of the requirements for a postgraduate degree from National Institute of Science Education and Research (NISER). I authorize NISER to lend this thesis to other institutions or individuals for the purpose of scholarly research.

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The thesis work reported in the thesis entitled "A pipeline for Venusian surface compositional study using VIRTIS data at near IR spectral windows with most surface contribution." was carried out under my supervision, at the School of Physical Sciences and Earth and Planetary Sciencesat NISER, Bhubaneswar, India.

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ABSTRACT

Venus remains largely unexplored due to its inhospitable conditions and thick cloud cover. The significance of analyzing surface composition and properties is immense, offering profound insights into the planet's geological history and potential for past habitability. Despite the global cloud shroud, specific near-infrared (NIR) spectral windows of nightside spectra allow for atmospheric transparency near the 1 μm region.

In particular, the spectral bands 1.02, 1.10, 1.18, 1.27, and 1.31 μm allow us to observe the surface with significantly lesser atmospheric hindrance in comparison to other bands. We use VIRTIS data from the Venus Express mission, i.e., the VIRTIS-M NIR spectral windows. The hyperspectral image contains geometric data including latitude, longitude, incidence, emergence, phase at footprint center, surface elevation, and slant distance. Data selection criteria include surface elevation data availability, zenith sun angle $\geq$ 98 degrees, and elevation angle $\leq$ 60 degrees. For pre-processing, each pixel is corrected for artifacts like spikes using polyfit techniques. Then each dataset is differentiated into background or space and planet points using threshold-based separation and DBSCAN with a high neighborhood radius. The average background spectrum is then subtracted from each planet spectrum for zero correction of each surface spectrum. This data is then extracted to be our OBSERVED SPECTRUM which gives us the radiance values: Radiance = $Wm^{-2}{\cdot}\mathrm{sr}^{-1}{\cdot}\mu\mathrm{m}^{-1}$.

These data need to be corrected for atmospheric scattering in order to retrieve the surface contribution. Our pipeline aims to perform atmospheric correction for the OBSERVED SPECTRUM using the Venus Global Reference Atmospheric Model (Venus-GRAM) software to get Temperature and Pressure profiles as well as the abundance profile for different gases, followed by Radiative Transfer Models (RTMs) simulating the atmospheric effect, including emission, absorption, and scattering. This allows us to have a SYNTHETIC ATMOSPHERIC SPECTRUM which is then removed from our observed data to get EXPECTED SURFACE SPECTRUM i.e., Emissivity from surface allowing us for surface compositional study.

With initial results from the pipeline, future work will focus on finalizing at-

mospheric corrections, defining uncertainties, and working on surface compositional analysis along with validation from Magellan data.

Contents

1 Introduction **1**
- 1.1 Background and Motivation . . . 1
- 1.2 The Space Race and Venus Exploration . . . 2
- 1.3 The Venus Express Mission . . . 3
- 1.4 VIRTIS Instrument Overview . . . 4
- 1.5 Research Objectives . . . 5

2 Methods **6**
- 2.1 Data Acquisition and Selection . . . 6
- 2.2 Pre-processing Pipeline/ . . . 7
  - 2.2.1 Spike Removal . . . 8
  - 2.2.2 Background Separation . . . 8
  - 2.2.3 Zero Correction . . . 9
- 2.3 Atmospheric Correction . . . 10
  - 2.3.1 Venus-GRAM . . . 10
  - 2.3.2 Radiative Transfer Model . . . 13
  - 2.3.3 Expected Surface Spectrum . . . 15
- 2.4 Validation Pipeline . . . 16

3 Results and Discussion **17**
- 3.1 Results . . . 17
- 3.2 Discussion . . . 18
  - 3.2.1 Limitations and Constraints . . . 19
  - 3.2.2 Methodological Advances . . . 19

4 Summary and Conclusions **22**
- 4.1 Summary . . . 22
- 4.2 Conclusions . . . 22
- 4.3 Future Work . . . 23

Appendix A **27**
- A.1 data . . . 27
- A.2 INITIALISATION . . . 27
- A.3 Data Correction Pre-processing . . . 28
  - A.3.1 Thermal Evolution Correction . . . 28
  - A.3.2 Bad Frame Identification and Correction . . . 28
  - A.3.3 Dead Pixel Correction . . . 28
  - A.3.4 Saturated Pixel Identification . . . 29
- A.4 Spectral Registration . . . 29
  - A.4.1 Subsystem Calibration, Florence . . . 29
  - A.4.2 Orsay Calibration . . . 29
  - A.4.3 Final Spectral Registration . . . 30
  - A.4.4 Wavelength Generation Routine . . . 30
- A.5 Radiometric Calibration . . . 30
  - A.5.1 Instrument Transfer Functions . . . 31
- A.6 Data Correction Post-processing . . . 31

A.6.1 Spike Removal . . . 31
A.6.2 Stripe Removal . . . 32
A.6.3 Spectral Filtering . . . 33
A.6.4 Spatial Filtering . . . 33
A.7 Finalization . . . 33
A.8 VIRTIS-M Spacecraft Alignment . . . 34
A.8.1 Alignment Measurement Strategy . . . 34
A.8.2 Observational Requirements . . . 34
A.8.3 SPC Orientation Procedure . . . 34
A.8.4 Coordination and Refinements . . . 35
A.9 VIRTIS-M Calibrated Data Structure . . . 35
A.10 VIRTIS Venus Express Geometry Files . . . 36
A.10.1 Geometric Data Contents . . . 36
A.10.2 Supplementary Planes . . . 36
A.10.3 Metadata and Auxiliary Parameters . . . 37
A.11 Scientific Objectives . . . 37
A.12 VIRTIS-M: Mapping Spectrometer . . . 40
A.12.1 Telescope . . . 40
A.12.2 Spectrometer . . . 40
A.12.3 Radiation Control and Filtering . . . 41
A.12.4 VIRTIS-M Filter Window . . . 41
A.13 VIRTIS Observation Cases at Venus . . . 42
A.13.1 Pericentre Observations . . . 42
A.13.2 Off-pericentre Observations . . . 43
A.13.3 Apocentre Observations . . . 43
A.13.4 Stellar Occultation (Three-axis Stabilization) . . . 43
A.13.5 Limb Sounding . . . 44
A.13.6 Surface Feature Tracking and Radiative Transfer . . . 44
A.13.7 Scientific Impact of Venus Express . . . 44
A.14 Synthetic spectra radiance . . . 44

List of Figures

1.1 Venus Express onboard instruments 3
1.2 VIRTIS-M 3D Data Cubes 4

2.1 Methodology flow-chart 6
2.2 .CAL files preparation 7
2.3 Spike removal using 2d polyfit. 8
2.4 Space, Planet and boundary seperation 8
2.5 Observed Spectra 9
2.6 Venus GRAM Example output 10
2.7 Inputs for RTM 13
2.8 Outputs from RTM 15
2.9 Ref. Plot : Crank et. al. 2007 15

List of Tables

3.1 Observed and Simulated Radiance Values ($Wm^{-2}sr^{-1}m^{-1}$) 17

A.1 Contents of VIRTIS Venus Express geometric data files 36

Chapter 1

Introduction

1.1 Background and Motivation

Venus, Earth's closest planetary neighbor, presents an intriguing challenge for planetary science due to its dense, reflective atmosphere. This atmosphere, predominantly composed of carbon dioxide (96.5%) and covered by thick sulfuric acid clouds, results in extreme conditions with surface pressures around 92 times that of Earth. These factors complicate remote sensing and data interpretation, yet understanding Venus's surface composition is crucial.

Analyzing Venus's diverse surface features—such as volcanic plains, tesserae, and mountain ranges—offers valuable insights into its volcanic and tectonic history. The surface preserves geological processes spanning billions of years, allowing scientists to reconstruct its geological timeline. Furthermore, Venus serves as a laboratory for studying planetary evolution under extreme conditions. By comparing Venus and Earth, researchers can explore the divergent evolutionary paths of terrestrial planets, understanding why Venus became inhospitable while Earth remains habitable. There is also significant interest in Venus's potential past habitability. Some hypotheses propose that Venus might have had liquid water and a temperate climate early in its history. Identifying specific minerals or hydrated materials on its surface could support or refute these claims, enhancing our knowledge of planetary habitability. Additionally, Venus provides a unique opportunity to study terrestrial planet formation and differentiation.

Comparing Venus with Earth and Mars offers critical insights into the processes

shaping inner solar system planets. Advancements in remote sensing technology and mission design have enabled unprecedented exploration of Venus. Spectroscopic observations within specific near-infrared (NIR) spectral windows—where the atmosphere is relatively transparent—are particularly valuable for studying surface composition. These observations help decipher the geology and evolution of this enigmatic planet, which continues to captivate scientific curiosity.

1.2 The Space Race and Venus Exploration

The Space Race, a significant period in the history of space exploration, greatly advanced our understanding of Venus. From the late 1950s, the competition between the United States and the Soviet Union led to remarkable progress in space technology and planetary science. For Venus exploration, this era laid the groundwork for later missions like the Venus Express.

The Soviet Union's Venera program was key in early Venus exploration. Venera 4 entered the Venusian atmosphere in 1967, providing the first in situ measurements and revealing the planet's extreme conditions. Following missions, such as Venera 9 and 10, achieved the first successful landings on Venus and returned the first images from its surface.

The United States' Mariner program was also instrumental. Mariner 2 successfully flew by Venus in 1962, confirming its harsh surface conditions and providing valuable data on its atmosphere and magnetic field.

These early missions, driven by the Space Race, established the foundation for our understanding of Venus. They demonstrated the challenges of exploring such a hostile environment and highlighted the need for advanced technologies and international collaboration.

This era's competitive yet collaborative nature set the stage for modern Venus exploration. Missions like the European Space Agency's Venus Express built upon this knowledge, carrying advanced instruments to study Venus's atmosphere, plasma environment, and surface properties in unprecedented detail.

1.3 The Venus Express Mission

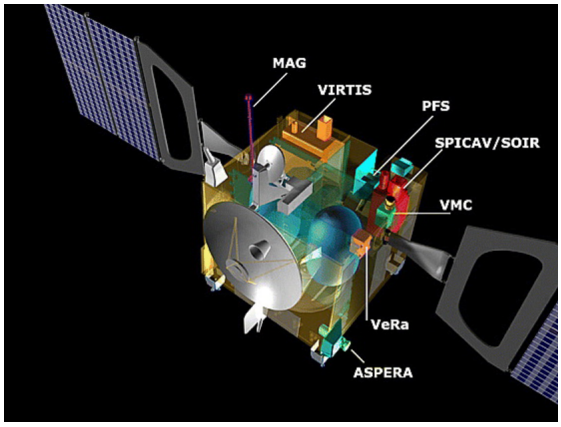


Figure 1.1: Venus Express onboard instruments

The European Space Agency's Venus Express (VEX) mission, operational from 2006 to 2014, marked a significant milestone in Venus exploration. As a successor to the Mars Express mission, VEX was designed to investigate the planet's atmosphere, plasma environment, and surface properties. Its highly elliptical polar orbit facilitated unique observational opportunities, balancing global coverage with detailed examinations of local features. Venus Express's elliptical orbit, ranging from 250 km at periapsis to 66,000 km at apoapsis, allowed for both close-up observations of the surface and atmosphere and broad surveys from afar. The spacecraft focused particularly on the southern hemisphere, utilizing its orbital geometry to gather high-resolution data and providing unprecedented insights into surface and atmospheric dynamics. By revisiting the same surface features under varying angles of incidence,

emergence, and phase, the mission enabled detailed studies of surface reflectance and emissivity.

1.4 VIRTIS Instrument Overview

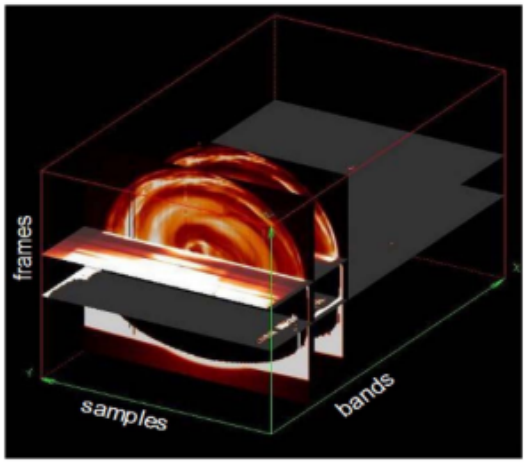


Figure 1.2: VIRTIS-M 3D Data Cubes

The Visible and Infrared Thermal Imaging Spectrometer (VIRTIS) onboard Venus Express was a versatile instrument designed for spectroscopic imaging of Venus's surface and atmosphere. Originally developed as a flight spare for the Rosetta mission, it was adapted for Venus exploration, covering wavelengths from 0.25 to 5.2 µm. VIRTIS featured:

- Spectral Range and Resolution: Spanning 1.0–5.2 μm with 432 spectral bands, it captured thermal emissions and reflected sunlight, ideal for nightside studies.
- Spatial Resolution: Collected data in hyperspectral cubes (256 spatial pixels per axis).
- Observation Modes: Optimized exposure times for both day and night observations.

VIRTIS included three channels: M-visible (0.25–1 µm), M-IR (1–5 µm) for spectral mapping, and H (2–5 µm) for high-resolution spectroscopy. These channels worked simultaneously or independently, providing spectral images combined with high-resolution spectra. Despite its advanced capabilities, VIRTIS faced challenges in extracting surface information due to atmospheric interference. Most of the planet's surface is obscured by intense scattering and absorption in the atmosphere, limiting observations to specific near-infrared spectral windows where these effects are minimal.

1.5 Research Objectives

This thesis focuses on utilizing VIRTIS-M data from Venus Express to analyze surface composition and emissivity with minimal atmospheric interference. The specific objectives are:

1. Data Preprocessing: Developing a robust pipeline to correct for instrument artifacts, identify surface-relevant spectra, and separate background contributions.

2. Atmospheric Correction: Using Venus-GRAM and radiative transfer models to simulate and remove the effects of atmospheric scattering, absorption, and emission from the spectra.

3. Validation and Analysis: Comparing corrected surface spectra with Magellan altimetry data to ensure consistency and identify correlations between surface emissivity and geological features.

4. Surface Composition Study: Interpreting the corrected spectra to deduce surface mineralogy and geological properties, contributing to the understanding of Venus's geological history.

Chapter 2

Methods

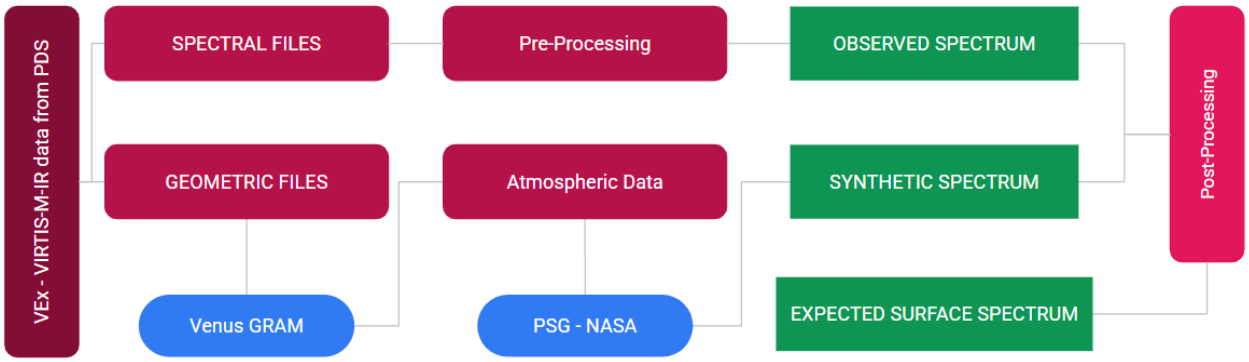


Figure 2.1: Methodology flow-chart

2.1 Data Acquisition and Selection

Our study utilizes nightside observations from the Visible and Infrared Thermal Imaging Spectrometer (VIRTIS-M-IR) onboard the Venus Express mission. These observations are critical for studying Venus's surface through near-infrared spectral windows, where atmospheric interference is minimal. The data were selected based on specific criteria to ensure quality and reliability:

- Surface Elevation Data Availability: Datasets were chosen only if surface elevation data were available for the corresponding geographic regions.
- Zenith Sun Angle $\geq$ 98 Degrees: Ensuring observations occur during the nightside phase to minimize solar interference.

- Elevation Angle ≤ 60 Degrees: Limiting observations to regions with low obliquity to reduce distortion and atmospheric path length effects.
- Geometric Data Quality Metrics: Inclusion of datasets with high-quality geometric information, including accurate latitude, longitude, and observation angles.
- Signal-to-Noise Ratio (SNR) Thresholds: Filtering datasets to retain only with those sufficient SNR for meaningful analysis.

These criteria were applied to the VIRTIS dataset to obtain a subset of high-quality data cubes for further analysis. Each data cube was uniquely identified by its orbit number and subset index.

2.2 Pre-processing Pipeline/

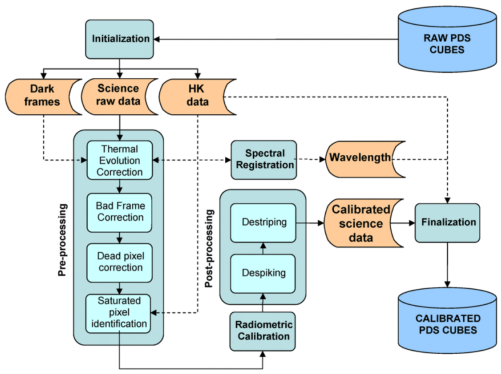


Figure 2.2: .CAL files preparation

The pre-processing pipeline was designed to handle the caliberated hyperspectral data cubes efficiently and prepare them for atmospheric correction and surface compositional analysis. The pipeline consisted of the following key steps:

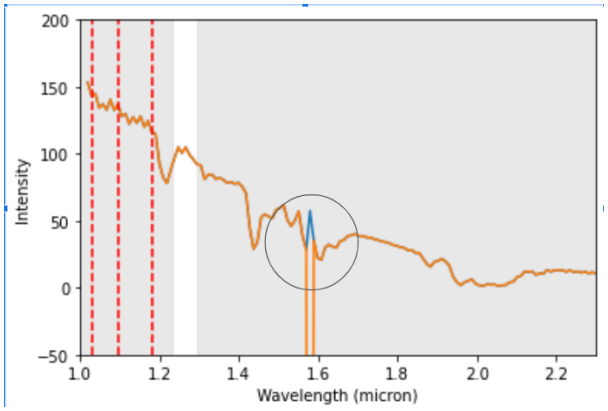


Figure 2.3: Spike removal using 2d polyfit.

2.2.1 Spike Removal

Instrumental and observational artifacts such as spikes in the spectral data were identified by their large values through threshold-based separators and removed using near-neighborhood corrections until the anomalies with respect to neighboring points were not greater than 10%. Then a second-degree polynomial was fit in place of the removed values to ensure consistency in data points.

2.2.2 Background Separation

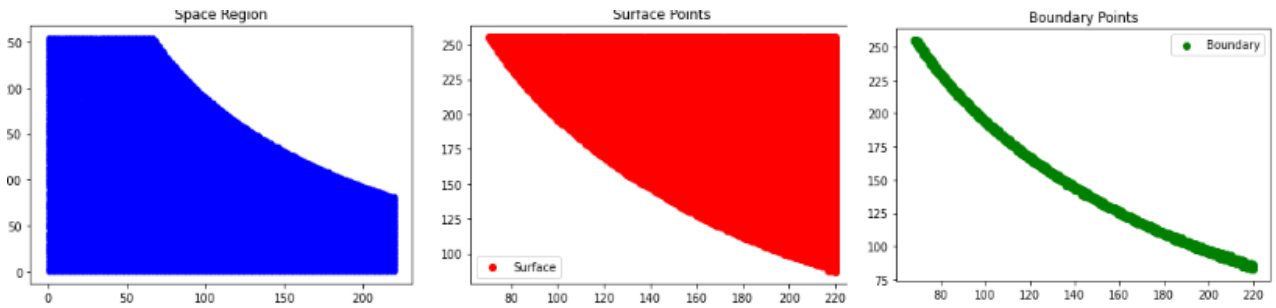


Figure 2.4: Space, Planet and boundary seperation

Separating planet-relevant data from background contributions is crucial for isolating the spectral signal from the Venusian surface. To ensure consistency across

datasets, the received spatial data was divided into space, boundary, and planet using spectral intensity thresholds based separation followed by DB Scan Clustering. This classified spectral data into planet and background points based on spatial and spectral proximity. Separation results were validated through statistical comparison of cluster properties, ensuring minimal overlap between background and planet data.

2.2.3 Zero Correction

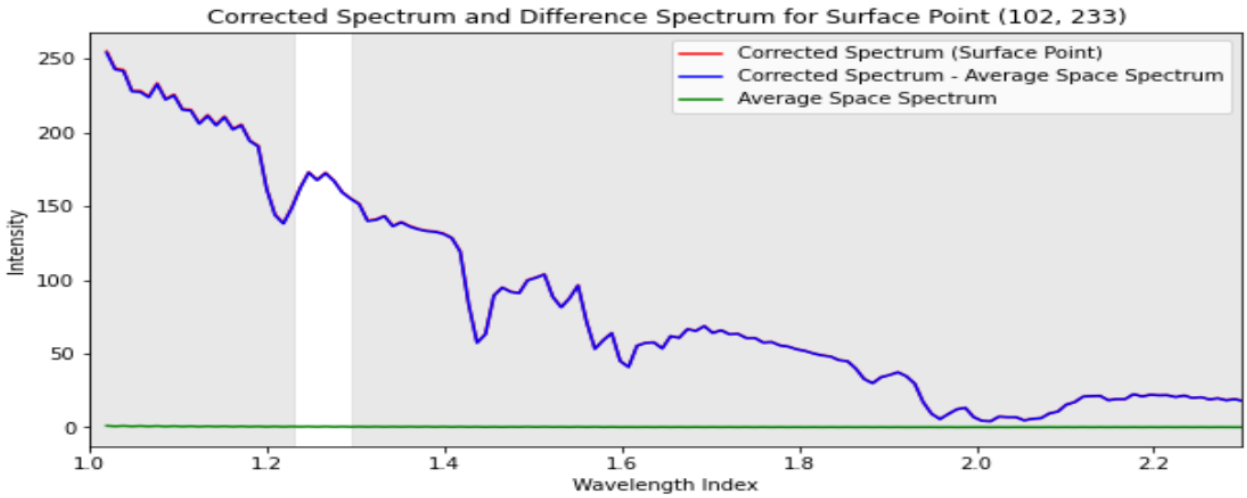


Figure 2.5: Observed Spectra

To normalize the surface spectra and eliminate systematic offsets, zero-point correction was applied:

- Background Spectrum Averaging: An average background spectrum was calculated from identified background points within each dataset.
- Surface Spectrum Normalization: The averaged background spectrum was subtracted from the surface spectra to remove residual background contributions.
- Validation of Correction Accuracy: Corrected spectra were validated by ensuring zero residual offset and consistency across overlapping datasets.

2.3 Atmospheric Correction

The atmospheric correction process aims to isolate the surface spectral radiance signal by accounting for atmospheric absorption, scattering, and emission. This is achieved using the Venus Global Reference Atmospheric Model (Venus-GRAM) software by NASA, which takes geometric information from VIRTIS as inputs to generate Pressure-Temperature profiles along with abundances for different gases. These profiles are then used in radiative transfer models to simulate the atmospheric effects, allowing us to generate a SYNTHETIC ATMOSPHERIC SPECTRUM, which is subtracted from the observed data to obtain the EXPECTED SURFACE SPECTRUM, i.e., the emissivity from the surface for surface compositional study.

2.3.1 Venus-GRAM

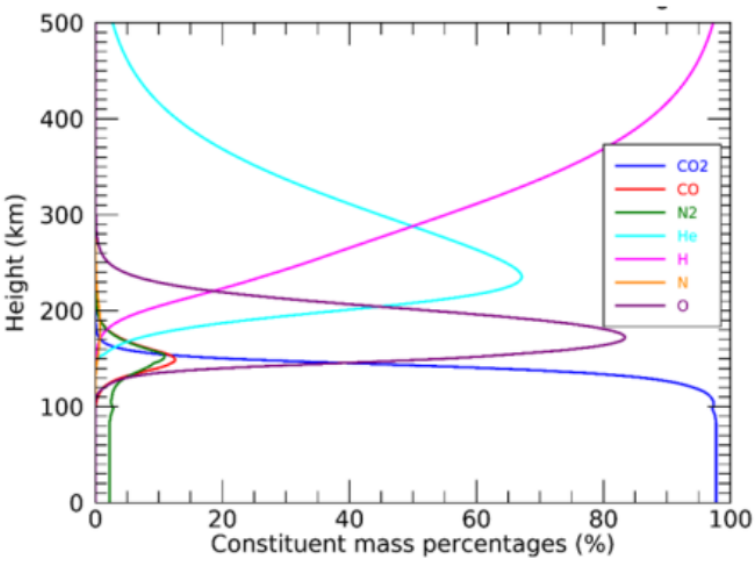


Figure 2.6: Venus GRAM Example output

Venus-GRAM uses the trajectory file provided along with VIRTIS-M-IR data, along with spatial coordinates and atmospheric modeling, to provide its output. At-

mospheric density, temperature, pressure, chemical composition, and winds as a function of height are characterized by the Venus International Reference Atmosphere (VIRA). The original version of VIRA in Venus-GRAM includes data from the Pioneer Venus orbiter and probe as well as the Venera probe, but it does not include a solid planet model or a high-resolution gravity model.

For the lower atmosphere (0 - 100 km), VIRA data depends on height and latitude. In the middle atmosphere (100 - 150 km), VIRA data depends on height and local true solar time. At upper altitudes (150 - 250 km), VIRA data depends on height and solar zenith angle. Venus-GRAM ensures smooth variation between height regions by averaging values at the two transition heights (100 km and 150 km). VIRA data incorporated in Venus-GRAM includes height, pressure, density, temperature, and number densities for CO2, N2, O, CO, He, N, and H. The Venus-GRAM thermosphere has been extended from the maximum VIRA height of 250 km to an altitude of 1000 km, using a model developed by the Marshall Space Flight Center (MSFC) based on VIRA conditions and constituents at 250 km as lower boundary values.

Monte Carlo iterations are employed in Venus-GRAM to simulate and analyze variations and uncertainties in input data, enabling a better understanding of possible outcomes. These iterations involve generating any number of independent trajectories that perturb key atmospheric parameters like density, speed of sound, and winds. By dispersing these parameters across multiple runs, Venus-GRAM helps capture the variability and probabilistic nature of the Venusian atmosphere.

Density Perturbations in Venus-GRAM

Venus-GRAM provides atmospheric density perturbations based on modeled turbulence and dynamic atmospheric effects. The perturbation magnitudes are estimated

using the following equations:

$$\rho' = \rho_0(1 + R' \cdot PF \cdot PU) \tag{2.1}$$

$$R' = e^{-SR} + X(1 - e^{-2S}) \tag{2.2}$$

Where:

- ρ' = perturbed atmospheric density value
- ρ_0 = mean atmospheric density
- R' = correlation factor for the current time step
- PF = modeled perturbation factor (typically height-dependent)
- PU = user-supplied perturbation multiplier
- S = relative displacement from the previous time step (considering NS, EW, vertical movement, and winds)
- R = correlation factor for the previous time step
- X = random number generator value

For small relative displacements, the new correlation factor remains close to its previous value ($R' \approx R$), whereas for large relative displacements, the correlation factor becomes effectively random ($R' \approx X$). This perturbation model enhances the realism of atmospheric density variability by incorporating stochastic and systematic influences from Venusian atmospheric dynamics.

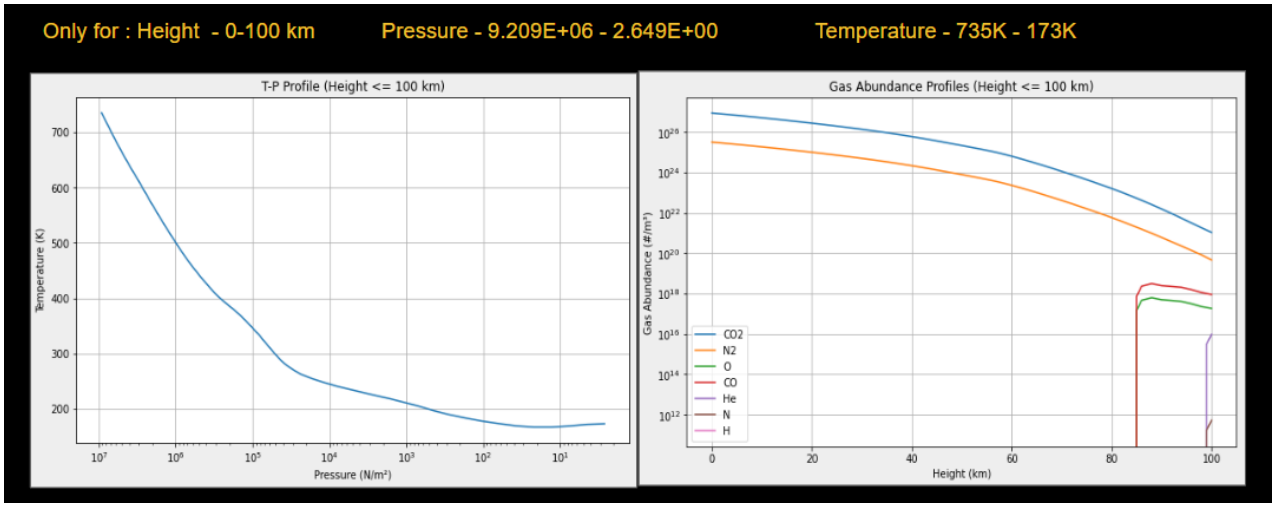


Figure 2.7: Inputs for RTM

2.3.2 Radiative Transfer Model

To accurately model Venus's atmospheric spectral radiance, a radiative transfer approach is employed, integrating pressure-temperature profiles and gas abundances extracted from Venus-GRAM. This methodology accounts for absorption, emission, and scattering processes within the dense Venusian atmosphere, including the 4 mode cloud opacity and Non-LTE interactions in Venus's upper atmosphere.

Model Overview

Radiative transfer calculations are performed to simulate the spectral radiance observed at different altitudes. The model incorporates:

- Vertical profiles of atmospheric pressure and temperature from Venus-GRAM.
- Gas species from Venus-GRAM, including CO2, N2, O2, CO, N, H, and He.
- Spectroscopic absorption coefficients computed using HITRAN and GEISA databases.
- Cloud scattering contributions, incorporating Venusian aerosol properties.

The core radiative transfer equation governing spectral radiance, $I(\nu, z)$, at a given frequency ν and altitude z is formulated as:

$$\frac{dI(\nu, z)}{dz} = -\kappa(\nu, z)I(\nu, z) + j(\nu, z) + S(\nu, z) \tag{2.3}$$

Where:

- $\kappa(\nu, z)$ represents the wavelength-dependent absorption coefficient.
- $j(\nu, z)$ is the source function accounting for thermal emission.
- $S(\nu, z)$ incorporates scattering effects from Venusian cloud aerosols and Rayleigh processes.

This RTM enables spectral fitting and correction of observed radiance, allowing for more accurate retrieval of surface emissivity and compositional variations on Venus. The model's performance is validated against VIRTIS-M spectral observations, demonstrating its robustness in distinguishing atmospheric and surface contributions.

Implementation using Venus-GRAM Profiles

The Venus-GRAM output provides temperature, pressure, and gas abundance profiles across various atmospheric layers. These parameters are interpolated and fed into the radiative transfer model along with geometric information to generate SYNTHETIC ATMOSPHERIC SPECTRA, i.e., surface flux at 1.02 µm. Monte Carlo simulations further refine the uncertainty estimates associated with radiative transfer by varying key atmospheric parameters to evaluate their impact on modeled spectral features.

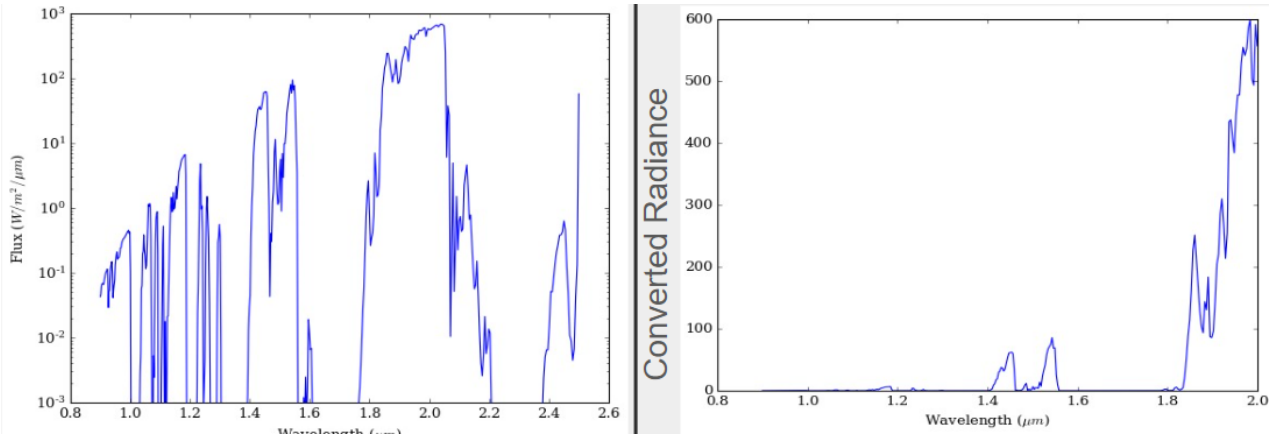


Figure 2.8: Outputs from RTM

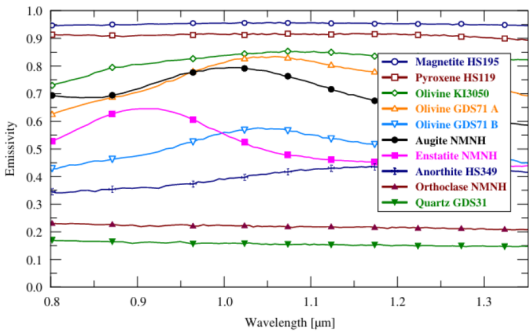


Figure 2.9: Ref. Plot : Crank et. al. 2007

2.3.3 Expected Surface Spectrum

Extracting surface-relevant information from the provided OBSERVED and SYNTHETIC radiances requires the following steps:

1. Simulating scattering from Venus dayside radiance and subtracting it from the nightside radiance to obtain OBSERVED SPECTRA.

2. Subtracting the SIMULATED ATMOSPHERIC SPECTRA (nightside radiance from atmosphere only) from the OBSERVED SPECTRA (nightside radiance from atmosphere and surface) to obtain the EXPECTED SURFACE SPECTRA (only surface contribution to OBSERVED SPECTRA) at the spectral window

of 1.02 µm.

3. (Optional) Converting the unit to emissivity from radiance or flux to get EXPECTED SURFACE EMISSIVITY (ESE).

4. Comparing the ESE with emissivity spectra or directly comparing the ESS with radiance ratios of potential Venus surface minerals to determine the EXPECTED SURFACE COMPOSITION.

5. Comparing the obtained composition against Magellan altimetry data to identify geological features suggesting the possibility of the expected composition.

Validation and Application The simulated spectral radiance is validated against observational data from VIRTIS-M to ensure consistency in retrieved atmospheric properties. By comparing modeled spectra with actual measurements, corrections for instrumental effects and retrieval biases can be systematically refined. This approach supports improved surface spectral retrievals by isolating atmospheric contributions.

2.4 Validation Pipeline

With atmospheric data from Venus Express now separated from the spectral data, a clearer surface spectrum is obtained. This spectrum enables the processing of geological features and composition of the Venusian surface. The following validation criteria are employed to ensure consistency in findings:

1. Processing the same dataset in parts to validate and emphasize geological properties.

2. Comparing with Magellan Altimetry Data.

3. Comparing across overlapping longitudes and latitudes for multiple datasets.

Chapter 3

Results and Discussion

3.1 Results

Table 3.1: Observed and Simulated Radiance Values ($Wm^{-2}sr^{-1}m^{-1}$)

Wavelength (µm)	**Observed Radiance**	**Simulated Atmospheric Radiance**
1.028	9.34291553159978e-08	0.03369908657
1.09471	1.5324624462800596e-05	0.03336392537
1.18019	6.446980776597632	0.07937622

The pre-processing pipeline for the VIRTIS dataset successfully obtained spectral data for the surface of Venus, free from background interference and correlated with their geological coordinates. This data is ready to be used for atmospheric corrections. The atmospheric properties retrieved, including Temperature-Pressure (T-P) and abundance profiles, are consistent with known atmospheric compositions with minimal deviations. The Venus-GRAM module was operated in both STANDALONE mode, where height parameters are provided to generate a trajectory file, and TRAJECTORY Mode, using the original trajectory files from the pre-processing pipeline. Almost consistent results were achieved, with the TRAJECTORY Mode data being used for further analysis. The Radiative Transfer Model (RTM) used in this study is a basic mode that ignores Non-LTE effects in the upper atmosphere and uses cloud opacity ratios as provided. This model utilizes the T-P profile from Venus-GRAM along with abundances of other gases to generate a synthetic atmospheric spectral radiance for a given latitude and longitude. Further development of the model is needed to extract true atmospheric data. Initial results from the RTM

show some promise, but the post-processing part is still under development. Based on spectral and geometric data from VIRTIS-M-IR from Venus Express, the P-T profile and gas abundances provided by Venus-GRAM, the radiative transfer model simulates gas-gas interactions along with cloud opacity factors, ignoring Non-LTE effects for now. Initial results as given in the table. Magellan altimetry data shows shield volcanic terrain, suggesting the presence of pyroxene-rich minerals with an emissivity of 0.85 ± 0.03, corresponding to a radiance of 5.2×10^{-4} W/m^2/sr/µm at 1.02 µm at 10N, 120E location. This is consistent with reference and previous knowledge. With ongoing RTM corrections, the aim is to improve the model for more reliable results. The validation pipeline remains a future aspect of the ongoing work. The spectral analysis of VIRTIS-M-IR data, after pre-processing and atmospheric correction, has revealed several findings related to Venus's surface properties and atmospheric effects. The study of surface emissivity patterns, particularly in the 1.02 µm spectral window, provides insights into the relationship between surface composition and topography, regional variations, and topographic correlations. The corrected VIRTIS data will be validated against altimetry data from the Magellan mission, providing a cross-check for accuracy and reliability. The comparison is expected to show consistency with regions and enhanced compositional details.

3.2 Discussion

The initial results from the pipeline demonstrate the potential for retrieving surface compositional information from VIRTIS data, despite the challenges posed by Venus's dense atmosphere. The atmospheric correction process, although still under development, has shown promising results in isolating the surface spectral signal. The integration of Venus-GRAM and radiative transfer models has been crucial in simu-

lating and removing atmospheric effects. However, there are several limitations and constraints that need to be considered.

3.2.1 Limitations and Constraints

- Atmospheric Model Assumptions: Venus-GRAM relies on simplified atmospheric profiles, which may not fully capture local variations in composition and temperature. Assumptions regarding scattering and absorption coefficients could introduce biases in corrected spectra.

- Temporal Coverage Gaps: The dataset does not provide continuous temporal coverage, limiting the ability to detect short-term surface or atmospheric changes. Seasonal and diurnal variations in atmospheric conditions remain under-explored.

- Spatial Resolution Constraints: The resolution of the VIRTIS-M-IR data limits the ability to study small-scale geological features. Comparisons with Magellan data, which has different resolution and coverage, introduce challenges in validation.

- Signal-to-Noise Limitations in Specific Bands: Certain spectral windows exhibit low signal-to-noise ratios, particularly in regions with dense cloud cover or low thermal contrast. This limits the ability to extract reliable surface information in these bands.

3.2.2 Methodological Advances

This study has introduced several methodological improvements that enhance the accuracy and reliability of Venus surface studies:

Refinement of Radiative Transfer Modeling

The development of a custom radiative transfer model tailored for Venus atmospheric studies marks a significant methodological improvement. Unlike previous models, this framework integrates Venus-GRAM atmospheric profiles with high-resolution spectroscopic databases (HITRAN, GEISA) to accurately simulate the interaction of CO2, N2, O, CO, H, and He in the dense Venusian atmosphere. Key enhancements include:

- Optimized spectral resolution for deep-atmosphere windows, particularly at 1.02–1.31 µm, allowing higher fidelity retrievals of surface emissivity.
- Improved scattering corrections, incorporating Rayleigh and aerosol scattering models to refine nightside spectral data interpretations.
- Dynamic perturbation framework, leveraging Venus-GRAM density fluctuations (ρ') to model turbulence-driven variations in radiative transfer calculations.

SPICE Integration for Precise Geometry Mapping

To enhance geospatial accuracy in surface flux retrieval, this study integrates SPICE kernels for spacecraft and planetary positioning. SPICE-based calculations enable precise corrections by aligning spectral data with corresponding surface latitudes, longitudes, and elevations.

- Custom CK and FK kernel selection ensures optimal spacecraft orientation mapping during Venus Express VIRTIS observations.
- Automated Python workflow using 'spiceypy' significantly reduces processing time, improving spectral cube alignment with Magellan radar altimetry data.

Surface Flux-to-Emissivity Conversion and Validation

Building upon our work as well as that of Poseidon RTM, we refine methods for surface emissivity derivation. Instead of using pre-existing RTM spectra directly, we employ a new correction strategy that subtracts modeled atmospheric contributions from observed spectra before flux-to-emissivity conversion.

- Adaptive correction model accounts for spectral contamination from atmospheric scattering and absorption variations.
- Comparative validation against Magellan radar maps ensures consistency in surface compositional assessments.

These methodological advances significantly improve the accuracy of retrieved Venus surface parameters, enabling high-confidence compositional interpretations.

Chapter 4

Summary and Conclusions

4.1 Summary

This thesis presents a comprehensive study of Venus's surface composition using VIRTIS-M data from the Venus Express mission. The research focuses on developing a robust data processing pipeline to correct for instrumental artifacts, atmospheric interference, and to retrieve surface spectral information. The key steps involved in this study include data acquisition and selection, pre-processing of hyperspectral data cubes, atmospheric correction using Venus-GRAM and radiative transfer models, and validation of the results against Magellan altimetry data.

The pre-processing pipeline efficiently handles the VIRTIS data cubes, correcting for spikes, separating background from planet data, and normalizing the surface spectra. The atmospheric correction process, although still under development, has shown promising results in isolating the surface spectral radiance signal. The integration of Venus-GRAM and radiative transfer models has been crucial in simulating and removing atmospheric effects. Initial results indicate the presence of pyroxene-rich minerals in specific regions, consistent with previous knowledge and expectations.

4.2 Conclusions

The study demonstrates the potential of VIRTIS data for Venus surface compositional analysis, despite the challenges posed by the planet's dense atmosphere. The methodological advances introduced in this research, including the refinement of radiative transfer modeling, SPICE integration for precise geometry mapping, and im-

proved surface flux-to-emissivity conversion, significantly enhance the accuracy and reliability of Venus surface studies. These improvements enable high-confidence compositional interpretations and provide a foundation for future research in this field.

However, there are limitations and constraints that need to be addressed in future work. The atmospheric correction process requires further refinement to account for local variations in atmospheric composition and temperature. Continuous temporal coverage of the dataset would enable the detection of short-term surface and atmospheric changes. Higher spatial resolution data would allow for the study of smaller geological features. Additionally, improving the signal-to-noise ratio in specific bands would enhance the reliability of surface information retrieval.

4.3 Future Work

- Finalizing Atmospheric Corrections: Refining the atmospheric correction pipeline to account for local variations in atmospheric composition and temperature.
- Defining Uncertainties: Quantifying the uncertainties associated with the retrieved surface spectral information.
- Surface Compositional Analysis: Expanding the surface compositional analysis to cover more regions and spectral windows.
- Validation Pipeline: Enhancing the validation pipeline with more comprehensive comparisons against Magellan data and other available datasets.
- Model Development: Continuously improving the radiative transfer model to better simulate atmospheric effects and surface contributions.

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Appendix A

A.1 data

VIRTIS-M: main purpose = spectral mapping visible and infrared at moderate spectral resolution Night side Surface mapping (0.9, 1.1, 1.18 µm) O2 emission (at 1.27 µm) Lower atmospheric sounding (H2O) Lightning search Day side UV cloud absorber signature / correlation with IR absorptions O2 emission (at 1.27 µm), also observed at day CO2 non LTE emissions (nadir and limb observations)

. CALIBRATION PIPELINE FOR VIRTIS-M The calibration pipeline for the VIRTIS-M channel is the same for both Visible and Infrared data and is performed locally in Rome. The calibration procedure can be divided in the following sections that are discussed in detail further below. 1. Initialisation 2. Data pre-processing 3. Spectral registration 4. Radiometric calibration 5. Data post-processing 6. Finalisation

A.2 INITIALISATION

The first phase of the calibration pipeline is simply an initialisation of the variables for the upcoming processes. Here a summarized list of the procedures performed during this first phase: 1.1. Constant definition 1.2. Parameter extraction 1.3. Raw file loading 1.4. Dark frame removal The dark frame removal is performed using the housekeeping information stored in the cube suffix which indicates the status of the shutter for each acquisition (see AD01 and RD18 for information about the Suffix structure). The acquisitions with shutter closed (darks) are identified and removed

from the cube

A.3 Data Correction Pre-processing

Several data correction procedures are applied to raw spectral data prior to radiometric conversion. These pre-processing techniques aim to mitigate instrumental and observational artifacts, ensuring accurate spectral retrieval.

A.3.1 Thermal Evolution Correction

Infrared and visible sensors exhibit thermally induced noise (Johnson noise), introducing temperature-dependent fluctuations in recorded signals. Additionally, infrared detectors capture not only radiation from the target but also background emissions from instrument components and the cold box. This background signal, referred to as "dark current," can lead to saturation in longer infrared wavelengths (4–5 µm), reducing signal-to-noise ratios and compromising spectral fidelity. Correcting for thermal noise is essential to recover the nominal spectral signal.

A.3.2 Bad Frame Identification and Correction

Occasionally, electronic anomalies within the sensor preamplifier result in corrupted frame acquisitions, affecting all spectral samples along the slit. These artifacts manifest as horizontal stripes in hyperspectral images, potentially interfering with scientific interpretation. Data processing pipelines systematically detect and mitigate these anomalies, ensuring restoration of affected frames.

A.3.3 Dead Pixel Correction

Long-term exposure to environmental conditions, particularly radiation, may progressively increase the fraction of non-responsive detector pixels. Continuous monitoring

of detector integrity allows for systematic identification and correction of dead pixels, preserving data quality across extended operational durations.

A.3.4 Saturated Pixel Identification

Infrared and visible detectors operate within specific dynamic ranges dictated by their electronic configurations. In the VIRTIS-M instrument, analog-to-digital conversions initially generate 16-bit pixel data, later formatted into 15-bit signed integers. Identifying and mitigating saturation ensures that spectral data remains within acceptable limits for subsequent analysis.

A.4 Spectral Registration

Spectral registration for both visible and infrared channels of VIRTIS-M was conducted through two on-ground calibration sessions—one in Florence (Italy) and another in Orsay (France). These sessions ensured accurate spectral mapping across operational conditions. The calibration methodologies, final computations, and results are summarized below.

A.4.1 Subsystem Calibration, Florence

The initial spectral profiling of VIRTIS-M was performed in Florence using a monochromator to scan across relevant wavelengths (see RD10 and RD11). These measurements were conducted at a cold case temperature of 135K, differing from nominal operational conditions in orbit.

A.4.2 Orsay Calibration

While Florence calibration provided spectral characteristics at a single temperature, the final spectral registration—relating band positions to wavelengths—was refined

at Orsay under varying thermal conditions (see RD05 in Section 1.5). This session utilized gas cells to account for temperature-dependent spectral shifts.

A.4.3 Final Spectral Registration

The spectral registration computations integrated results from Orsay to refine initial fits obtained in Florence. Specifically, the final temperature-dependent slope and intercept corrections were derived, ensuring accurate spectral characterization across operational conditions.

A.4.4 Wavelength Generation Routine

The spectral calibration results were compiled into the WAVELENGTH_GEN routine ("wavelength_gen.pro"), included in the calibration package. This routine systematically incorporates the computed parameters—slope and intercept—to generate wavelengths for both visible and infrared channels with precision.

A.5 Radiometric Calibration

Raw spectral data cubes record sensor-detected signals in Digital Numbers (DN). To obtain physical units of spectral radiance, $\mathrm{Rad}(W\,m^{-2}\,sr^{-1}\,\mu m^{-1})$, the following conversion is applied:

$$\mathrm{Rad}(b, s, l) = \frac{\mathrm{DN}(b, s, l)}{t \cdot R(b, s)} \tag{A.1}$$

where b, s, l represent spectral band, sample, and line location within the original data cube; t denotes the integration time (in seconds) as recorded in the PDS file header for VIS and IR channels; and $R(b, s)$ corresponds to the 2D responsivity matrix for VIS and IR channels. For high-resolution acquisitions (432 bands $\times$ 256 samples), this transformation is directly applied. In lower-resolution modes, spectral

tables and responsivity matrices undergo interpolation in accordance with binning parameters.

A.5.1 Instrument Transfer Functions

Spectral processing for Visible and Infrared channels differs primarily in the application of distinct Instrument Transfer Functions (ITFs). Each ITF is stored as a binary file, structured as a two-dimensional matrix representing sensor responsivity across spectral elements. The responsivity unit is given by:

$$\left(\frac{m^2 \cdot sr \cdot \mu m}{W \cdot s}\right) \tag{A.2}$$

These ITFs ensure precise radiometric calibration, enabling accurate spectral interpretation for Venus surface studies.

A.6 Data Correction Post-processing

Certain data correction procedures must be performed after radiometric conversion to ensure that instrumental effects do not interfere with scientific interpretations. These post-processing techniques refine spectral data by eliminating anomalies.

A.6.1 Spike Removal

Single-event artifacts, such as cosmic ray hits or heavy ion interactions, introduce spurious spikes in spectral frames. While isolated pixel events can be easily corrected, extended anomalies affecting adjacent pixels require systematic removal. The calibration pipeline employs the `DESPIKE` routine (`despike.pro`) to detect and mitigate such effects.

Algorithm Overview:

1. A pixel is classified as a spike if its value deviates beyond the median of its local neighborhood by a predefined margin.

2. The local neighborhood consists of the target pixel and its eight adjacent pixels within the sample-band frame.

3. The valid margin is determined using the standard deviation of the local region, scaled by a threshold (typically 3.0).

4. Standard deviation is estimated using the difference between the second highest and second lowest pixel values, divided by two.

5. If the data cube contains NaN values, spike detection is suppressed for pixels neighboring NaN locations.

A.6.2 Stripe Removal

Vertical stripe artifacts arise in VIRTIS observations when certain pixels exhibit varying responsivity along the slit, independent of instrumental transfer functions. To address this, the calibration pipeline applies the `DESTRIPING_SELECTIVE` routine (`destriping_selective.pro`).

Algorithm Overview:

1. Identify valid spectral data, excluding deep space, saturated pixels, and other anomalies.

2. Construct a mask for each spectral band to define valid regions, avoiding dead pixels and artifacts.

3. Compute the column-wise average within the masked region for each spectral band.

4. Apply correction factors to each column, adjusting based on deviations from neighboring column averages. Corrections typically range between 2–3% of measured radiance.

Results: The stripe correction process significantly improves spectral quality by minimizing vertical artifacts while preserving spectral integrity.

A.6.3 Spectral Filtering

The current version of the calibration process does not include spectral filtering. Future refinements may integrate spectral correction algorithms to enhance spectral fidelity by mitigating instrumental artifacts.

A.6.4 Spatial Filtering

Spatial filtering is not applied in the present calibration pipeline. Advanced spatial correction techniques may be considered in subsequent iterations to improve spatial uniformity and suppress regional anomalies.

A.7 Finalization

The finalization phase encompasses all post-calibration procedures required to ensure completeness of the processed data. This stage includes:

- Structuring the calibrated data cube in its definitive format.
- Generating additional metadata and supplementary diagnostic information.
- Creating output files compatible with validation and scientific analysis frameworks.

These steps mark the transition of raw calibrated data into its finalized form, ready for subsequent spectral analysis and geological interpretation.

A.8 VIRTIS-M Spacecraft Alignment

The spacecraft alignment measurement for VIRTIS-M follows a procedure similar to that of the Rosetta mission. However, due to limited accessibility to the VIRTIS reference cube during the Venus Express ground alignment campaign, this measurement plays a crucial role in confirming the alignment offset necessary for VIRTIS-H calibration. To ensure accuracy, the alignment measurement must be conducted well in advance of the VIRTIS-H alignment phase.

A.8.1 Alignment Measurement Strategy

Given the short cruise duration to Venus and restricted recovery windows, alignment measurements are prioritized immediately after the first activation of VIRTIS during NEV activity. The proposed target for alignment validation is the Earth-Moon system, serving both scientific and calibration purposes. This selection allows for direct comparison with similar alignment campaigns conducted during the Rosetta mission.

A.8.2 Observational Requirements

To achieve precise alignment characterization, the spacecraft will be pointed such that the target is centered on the boresight of VIRTIS-M. High-resolution spectral and spatial imaging will be conducted with VIRTIS-M operating independently. Additionally, a secondary spacecraft re-pointing maneuver is required to eliminate ambiguity in defining the orientation plane, rather than solely the boresight axis.

A.8.3 SPC Orientation Procedure

Two spacecraft orientations will be executed, differing by an angular displacement of approximately 1° (either along-track or across-track, to be determined). These adjustments ensure a comprehensive understanding of the alignment parameters.

A.8.4 Coordination and Refinements

Finalization of the detailed alignment procedure remains pending and requires collaborative validation with VSOC and confirmation by VMOC to ensure procedural consistency and optimal calibration accuracy.

A.9 VIRTIS-M Calibrated Data Structure

The calibrated VIRTIS-M data preserves the original 3D format of raw observations, ensuring consistency in spectral mapping and geometric correlations. The output structure comprises:

- **result.label**: Label of the PDS file.
- **result.table**: A 3D array storing spectral references per matrix spectel:
  - result.table(*,*,0) = Wavelength.
  - result.table(*,*,1) = Bandwidth (FWHM).
  - result.table(*,*,2) = Radiance uncertainty (1-sigma).
- **result.qube_name**: A two-string array indicating the stored cube quantity and unit.
- **result.qube_dim**: A three-element array defining cube dimensions.
- **result.qube**: 3D core data cube (floating-point values), with 3 dimensions .
- **result.suf_name**: List of suffix parameters (SCET).
- **result.suf_dim**: A two-element array specifying suffix dimensions.
- **result.suffix**: SCET values recorded for each frame, formatted as (3, #frames).

A.10 VIRTIS Venus Express Geometry Files

The VIRTIS Venus Express dataset includes geometry files containing projection parameters for Venus observations, viewing angles, surface elevation, and related spatial attributes. The geometric parameters are stored within Qube objects without sideplanes, with conversion coefficients provided in `result.qube_coeff` to transform values into standard units (e.g., distances in km, angles in degrees, local time in Venus hours).

A.10.1 Geometric Data Contents

The following table outlines the parameters stored in the geometric data qubes:

Plane #	Parameter Description
1–4	Longitudes of four pixel footprint corners
5–8	Latitudes of four pixel footprint corners
9–10	Center longitude and latitude on surface ellipsoid
11–13	Incidence, emergence, and phase angles at footprint center
14	Surface elevation (averaged footprint corners)
15	Slant distance (line-of-sight to surface ellipsoid at pixel center)
16	Local time at footprint center
17–20	Longitudes of four pixel footprint corners at cloud layer (60km)
21–24	Latitudes of four footprint corners at cloud layer
25–26	Longitudes and latitudes of pixel footprint center on cloud layer
27–29	Incidence, emergence, and phase angles at cloud layer footprint
30	Surface elevation at vertical intercept of cloud layer
31–32	Right ascension and declination of pointing direction

Table A.1: Contents of VIRTIS Venus Express geometric data files

A.10.2 Supplementary Planes

For VIRTIS-M, an additional common plane (#33) is included:

- Stores ten scalar quantities along spatial frame dimensions.
- Remaining values initialized to zero.

A.10.3 Metadata and Auxiliary Parameters

- Original data SCET timestamps from telemetry (integer and fractional parts).
- Encoded UTC using SPICE computations.
- Subspacecraft coordinates (longitude, latitude).
- Mirror angle parameters transformed into sine and cosine values from housekeeping (HK) data.
- Sun direction properties:
  - Angle between Sun direction and spacecraft Z-axis.
  - Azimuth of Sun direction within the instrument XY-plane.

A.11 Scientific Objectives

Observations of Venus's nightside in the near-infrared (NIR) from Earth, first conducted by Allen and Crawford (1984), provided initial evidence of deep atmospheric sounding. Subsequent high-resolution spectroscopic measurements (Bézard et al., 1990) further refined the understanding of atmospheric composition, including water vapor and carbon monoxide, which are critical for thermochemical equilibrium studies (Kamp et al., 1988).

The first spacecraft-based NIR imaging spectrometry of Venus's nightside was carried out by the Galileo Near-Infrared Mapping Spectrometer (NIMS) in 1990 (Carlson et al., 1991) and Cassini's Visual and Infrared Mapping Spectrometer (VIMS) in 1999 (Baines et al., 2000). These flybys demonstrated the potential of this technique for Venus studies; however, their short durations limited investigations, particularly regarding cloud evolution.

The VIRTIS instrument aboard Venus Express presents a unique opportunity to extend these investigations through continuous observations of both the dayside and nightside in the infrared and visible spectrum. The scientific objectives of VIRTIS observations include:

- **Lower Atmosphere Composition:** Mapping variations in CO, OCS, SO_2, and H_2O below the cloud layers using nightside observations. The 2.3 μm window, captured at high resolution with VIRTIS-H, enables precise measurement of minor species, while H_2O variations can be mapped with VIRTIS-M (Drossart et al., 1993). These observations provide insights into volcanic and meteorological activities, particularly in relation to sulfur compound emissions.

- **Cloud Structure and Composition:** Investigating spatial and temporal variability in cloud optical properties on both the dayside and nightside. VIRTIS's spectral coverage enables detailed characterization of scattering properties and constraints on cloud layers.

- **Cloud Tracking and Wind Field Analysis:** Tracking cloud dynamics in the ultraviolet ($\sim$70 km altitude, dayside) and infrared ($\sim$50 km altitude, nightside). Correlating UV and IR observations across different timescales aids in reconstructing wind field variations up to 70 km altitude.

- **Temperature Field Retrieval and Zonal Wind Determination:** Utilizing thermal emission measurements in the 4–5 μm range to derive Venus's temperature field at 60–100 km altitude (Roos et al., 1993). The retrieved thermal structure supports zonal wind estimations through cyclostrophic approximation.

- **Lightning Search:** Investigating transient lightning phenomena in the Venusian atmosphere, though its occurrence remains uncertain. Detecting lightning would provide key insights into atmospheric electrical activity.

- **Mesospheric Sounding:** Studying the transition region between the troposphere and thermosphere through:

  - Non-local thermodynamic equilibrium (non-LTE) O_2 emissions at 1.27 μm (95–110 km; Drossart et al., 1993).

  - CO_2 fluorescence emissions at 4.3 μm (>80 km), aiding upper-atmosphere physics studies (Roldan et al., 2000).

  - Limb observations of CO and CO_2 for vertical structure characterization at altitudes exceeding 60 km.

- **Surface-Atmosphere Interaction Studies:** Searching for spatial and temporal variations related to atmospheric dynamics, meteorology, and volcanism. Global and regional-scale observations aim to identify correlations between distinct physical processes.

- **Surface Temperature Mapping and Volcanic Hotspot Detection:** Investigating thermal anomalies that may indicate volcanic activity. While Venus's atmosphere complicates lava detection, temperature variations could serve as indirect indicators.

- **Seismic Wave and Gravity Wave Investigations:** Searching for acoustic wave propagation through mesospheric pressure/temperature variations in the CO_2 4.3 μm band (Artru et al., 2001). Understanding gravity wave activity contributes to knowledge of atmospheric circulation.

These objectives, pursued during the Venus Express mission, provide a comprehensive framework for advancing Venus science.

A.12 VIRTIS-M: Mapping Spectrometer

VIRTIS-M is the mapping spectrometer component of the VIRTIS instrument aboard Venus Express. It comprises the M-optical head within the optical module (OM) and the PEM-M proximity electronics, which reside in the spacecraft's thermally controlled environment. The optical system integrates both visible and infrared channels into a single optical axis, ensuring co-aligned spectral observations.

A.12.1 Telescope

The VIRTIS-M telescope employs a Shafer-type optical design, combining an inverted Burch telescope with an Offner relay. The Offner relay transforms the curved, anastigmatic virtual image produced by the inverted telescope into a flat, real image while preserving its anastigmatic quality.

Coma aberrations are mitigated by placing the aperture stop near the primary mirror's center of curvature, rendering the system monocentric. This configuration enables diffraction-limited performance across an appreciable spectrum and ensures high optical quality along the vertical slit direction. The horizontal field is achieved by rotating the telescope mirror around an axis parallel to the Y-axis of the spacecraft.

A.12.2 Spectrometer

Unlike conventional imaging spectrometers used in Cassini's VIMS and Mars Express's OMEGA (Bibring et al., 2004), VIRTIS-M's Offner spectrometer eliminates redundant optical components, such as collimators and beam splitters, thereby simplifying fabrication and minimizing volume and mass. This design requires perfect

optical matching between the telescope and spectrometer, including telecentric properties, ensuring optimal performance.

The spectrometer grating implements a dual-zone approach with distinct groove densities (Piccioni et al., 2000). Holographically recorded and ion-etched grooves create spectral beam splitting tailored for each field-of-view angle:

- **Visible channel (inner 30% of pupil area):** Higher groove density enhances spectral resolution from ultraviolet to near-infrared, enabling fine spectral mapping with a smaller point spread function.
- **Infrared channel (outer 70% of pupil area):** Lower spectral resolution is accepted due to modulation transfer function (MTF) constraints arising from visible-channel pupil obscuration.

A.12.3 Radiation Control and Filtering

To mitigate background photon emissions beyond 4.2 μm, infrared focal plane arrays (IRFPAs) are shielded by spectral blocking filters with 5.1 μm cut-off wavelengths. A dedicated long-pass filter further prevents overlapping higher-order diffraction from visible and infrared channels.

Since background radiation in the 4.2–5.1 μm range cannot be filtered within the collected spectral bands, the spectrometer is actively cooled to temperatures below 150K to minimize thermal interference.

A.12.4 VIRTIS-M Filter Window

The VIRTIS-M filter window plays a crucial role in preventing the superimposition of higher diffraction orders from the spectrometer grating while minimizing background thermal radiation caused by the instrument housing. To optimize spectral perfor-

mance, the window is covered with an anti-reflection coating specifically designed for the 1–5.1 μm range.

Additionally, five segment filters are integrated onto the window, providing well-defined spectral bandpasses:

- **Region 1:** 0.9–1.6 μm
- **Region 2:** 1.2–2.5 μm
- **Region 3:** 2.4–3.75 μm
- **Region 4:** 3.6–4.4 μm
- **Region 5:** 4.3–5.0 μm

These segmented filters enhance spectral discrimination by efficiently blocking unwanted wavelengths while preserving the integrity of the observed signals. The implementation of this filtering mechanism is essential for optimizing data retrieval, particularly for Venus's infrared spectral studies.

A.13 VIRTIS Observation Cases at Venus

VIRTIS employs multiple observation strategies to study Venus's atmosphere, surface, and meteorological phenomena. These modes facilitate spectral mapping, dynamic monitoring, and structural analysis.

A.13.1 Pericentre Observations

Pericentre observations provide statistical coverage of Venus's surface across different regions, both during the day and night. The scanning mirror, used in either fixed or scanning mode, enhances coverage but complicates image reconstruction due to the high velocity of the spacecraft.

Pixel dimensions are defined as the linear size of the VIRTIS-M high-resolution spatial pixel (250 μrad in angle) on Venus's surface at the slit center. Due to short dwell times, either elongated pixels or missing data regions ("holes") may occur when the scanning mirror is engaged. Observations of identical regions at varying phase angles (feature tracking) assist in characterizing the scattering particle phase function (H_2SO_4), crucial for radiative transfer modeling and mean particle size estimation.

A.13.2 Off-pericentre Observations

Spectral mapping at optimal spatial resolution occurs during off-pericentre observations. Internal mirror motion synchronized with orbital motion enables image cube reconstruction, requiring an altitude above 12,000 km. This mode is extensively used for combined spectroscopic and mapping objectives.

A.13.3 Apocentre Observations

Apocentre observations enable full-disc imaging of the southern hemisphere using a 3 $\times$ 3 mosaic technique. This mode supports global meteorological studies, atmospheric rotation analysis, cloud property mapping, wind dynamics, and temperature field monitoring. Repeated observations across multiple orbits help characterize long-term atmospheric behavior and are also valuable for public outreach imagery.

A.13.4 Stellar Occultation (Three-axis Stabilization)

VIRTIS collaborates with SPICAV in this mode to retrieve upper atmospheric composition through stellar flux absorption studies. This technique identifies minor atmospheric constituents and extracts CO_2 absorption profiles, aiding upper-atmosphere structure analysis. Geometrically, this mode aligns closely with limb sounding observations.

A.13.5 Limb Sounding

VIRTIS's unique capability to image in the 4.3 μm CO_2 band enables mesospheric emission studies. Limb observations provide vertical resolution, allowing extended atmospheric path probing at and above cloud tops, reaching altitudes up to 115 km in the thermosphere. This method complements limb observations conducted by Mars Express/OMEGA.

A.13.6 Surface Feature Tracking and Radiative Transfer

High-priority observations target select Venusian surface features, though the planet's slow rotation limits opportunities. Nadir and near-nadir viewing, combined with feature-tracking pointing techniques, facilitate radiance studies across multiple phase angles. These observations refine radiative transfer models and atmospheric parameter retrieval.

A.13.7 Scientific Impact of Venus Express

Venus Express marks a crucial step in the systematic exploration of Venus. Although it will not resolve all mysteries surrounding the dense Venusian atmosphere, VIRTIS significantly enhances the scientific return of the mission. It is anticipated that findings from Venus Express will inspire future planetary exploration efforts.

A.14 Synthetic spectra radiance

VIRTIS-M-IR measurements cover the spectral region 1–5 mm. Fig. 1 shows typical radiance spectra as a function of wavelength for both the dayside and the nightside of Venus for different observer and illumination geometries, detector exposure times and for different environmental conditions on the planet. The huge difference in the dayside measurements is mainly due to different solar illumination (incidence angles

of 35.71 and 86.81, respectively). Fig. 2 explains where the very different features result from. Computational details of synthetic radiance spectra are described later on in Section 3. Dayside signatures (upper thick line) below 3.5 mm originate from scattering of the intense sun light in the upper cloud layer. Above 3.5 mm, there is an increase in influence of thermal radiation from the atmosphere. This is illustrated by the dashed line, which reflects the contribution of the pure solar component to the dayside signatures. Dayside measurements are especially useful below 2.5 mm to measure, for example, cloud top altitudes from the CO2 feature located at 1.6 mm(Ignatiev et al., 2009). Other applications include scale height, scattering phase function, and H2O concentration retrieval. This is the reason why most VIRTIS dayside measurements are configured to yield a good signal-to noise ratio in that spectral region using very short detector exposure times. On the other hand, this leads to very noisy spectra above 2.5 mm. Nightside spectra (lower thick line) are mainly formed by thermal emissions from different atmospheric layers and the surface that are modified by subsequent cloud scattering.